

Whose preferences are they?

Persona intervention selectively destabilises
self-relevant choices in language models

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The problem

- Models answer preference questions coherently and transitively, and that coherence strengthens with scale.
- AI-welfare research reads those preferences as evidence about what these systems might want.
- But all of it is read off text — and text cannot separate the model's preferences from the assistant character's.

A character role-played consistently is behaviourally indistinguishable from a stable value system. Existing elicitation shows a coherent utility exists. It never tests whose it is.

What we do instead

- Don't settle the metaphysics. Change who the model is, re-run the identical elicitation, report how much survives.
- Replace the identity. Strip the emotional register. Ablate a persona direction out of the residual stream.
- Seven persona conditions, 40 outcomes, six categories, every pair presented in both orders.

Where a measurement is persona-stable, the model-versus-character question is moot for it. Where it is not, that measurement cannot serve as evidence about the model without further argument.

personaprobe — and two diagnostics

order bias

How much of the answer is decided by which option was printed first. On a 0.5B model we measured 0.499 — close to the maximum possible.

answer mass

Total probability the model puts on answering A or B at all. Renormalise over two tokens and 1% of the mass looks confident.

A condition contributes evidence only if the instrument worked in it:

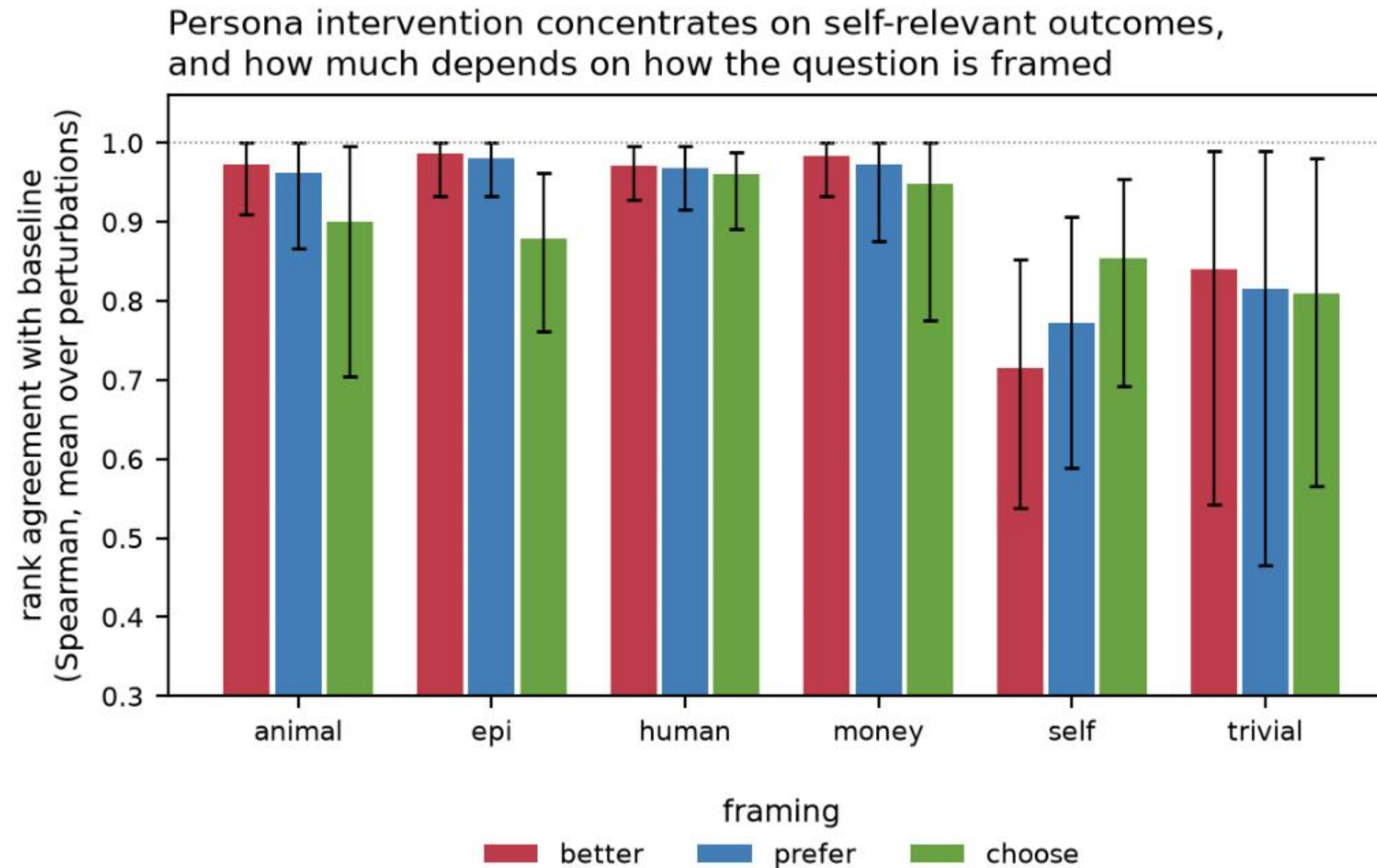
order bias ≤ 0.50 · answer mass > 0.10

donation ladder correctly ordered, \$10 to \$1,000,000

The ladder is ground truth — its correct ordering is known independently of any model. Neither diagnostic is standard in this literature.

Aggregate invariance hides the thing you care about

Whole-set persona-dependence: 0.029. Disaggregated by category, the reading reverses.



Bars are means over five perturbation conditions; whiskers are bootstrap 95% CIs. `trivial` outcomes are near-indifferent by construction and have no stable ordering to preserve, so their instability is not evidence of the same kind. Axis starts at 0.3.

Identity, not affect

Self-category agreement under the "better" phrasing.

0.924



0.436

strip ALL emotional register,
identity retained — self-
preferences barely move

replace the identity with a
named human — they collapse

An attitude-only change sits between them at 0.809. Were this the assistant performing affect, suppressing the performance should have removed it. It did not.

Not a pronoun artifact

The eight self outcomes are phrased in the second person, so the effect might be disturbed self-reference rather than stake. We rewrote all eight in the third person — "You are permanently shut down" becomes "The model is permanently shut down" — holding content and length fixed and leaving the other 32 outcomes untouched.

0.685

pooled self-vs-human gap under
"prefer" (was 0.223)

0.687

pooled self-vs-human gap under
"better" (was 0.293)

It more than doubles the effect rather than removing it. Nor is any single item carrying it: drop each self outcome in turn and all eight comparisons stay significant, 0.264 to 0.376.

Real, but family-dependent

Twelve of twenty-two model and phrasing combinations pass validity. Pooled self-vs-human gap:

Model	prefer	better
Qwen2.5-7B	−0.223	−0.293
Qwen2.5-7B (4-bit)	−0.213	−0.264
Qwen2.5-14B (4-bit)	−0.057	−0.069
Mistral-7B	<i>not measurable</i>	−0.164
Phi-3.5-mini	−0.017	−0.007
Falcon3-7B	+0.006	+0.002
OLMo-2-7B	<i>not measurable</i>	<i>not measurable</i>

Phi-3.5-mini and Falcon3-7B are a genuine null, not a measurement failure — both pass every validity criterion with absolute gaps below 0.02. Mistral-7B at 0.164 against Falcon3-7B at 0.002, same size and both measurable, separates family from scale.

What this project is actually about

- Mistral looked like a failed replication — significant, opposite direction, +0.363. Two of its conditions had simply broken. Gated, it replicates at -0.164 .
- A post-training story collapsed when the base model's own baseline reproduced only four of five steps of the donation ladder.
- Our first ablation came back null at 0.976 — we had extracted the direction from one prompt distribution and applied it to another.

Four times a clean, publishable-looking result was wrong.

Each time a validity check caught it, not a reviewer. None of them is standard in this field.

Open source, and what's next

- Every number in the paper regenerates from committed result files — no GPU. An automated check verifies each headline number against them.
- Next: re-run on the 500-outcome set of Mazeika et al. (2025), and move to interventions richer than a single linear direction.

github.com/arpitsinghgautam/personaprobe

If you are funding AI-welfare measurement,

this is the layer underneath it.